ModelIt!

Teacher's Guide

Survival of the Fittest: Can You Outrun Natural Selection?

How Environmental Pressure and Genetic Diversity Shape Who Survives

Quick Reference

Grade Level	Grade 08L2
NGSS Standard	MS-LS4-4, MS-LS4-6
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-LS4-4, MS-LS4-6

Construct an explanation based on evidence that describes how genetic variations of traits in a population increase some individuals' probability of surviving and reproducing in a specific environment. Use mathematical representations to support explanations of how natural selection may lead to increases and decreases of specific traits in populations over time.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model to demonstrate how natural selection changes trait frequencies in a population over multiple generations.
Disciplinary Core Idea	**LS4.B Natural Selection** — Natural selection leads to the predominance of certain traits in a population and the suppression of others based on environmental conditions.
Crosscutting Concept	**Cause and Effect** — Students identify that environmental pressure CAUSES differential survival, which leads to the EFFECT of changed trait frequencies across generations.

CAST Testing Alignment

MULTIPLE CHOICE:

A population of lizards lives in a region where temperatures are rising. Only 15% of the population carries a heat-tolerance gene. After 10 generations of intense heat, 80% of survivors carry the gene. Which best explains this shift?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain why a population with high genetic diversity is more likely to survive a sudden environmental change than a population with low diversity. Reference specific components and relationships from your simulation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Natural selection is one of the key mechanisms of evolution, first described by Charles Darwin and Alfred Russel Wallace. It explains how populations change over time as environmental pressures favor certain inherited traits over others. Understanding natural selection is critical for conservation biology, agriculture, and medicine.

The Three Requirements for Natural Selection

Natural selection requires three conditions: variation (individuals differ in their traits), heritability (traits are passed from parents to offspring), and differential survival (some traits provide better chances of surviving and reproducing in the current environment). When all three are present, populations evolve.

Genetic Diversity as a Survival Tool

A population's genetic diversity is its toolkit for facing change. Large, diverse populations contain many trait variations, increasing the odds that some individuals will have traits suited to new conditions. Small, inbred populations with low diversity are vulnerable because they may lack the genetic 'raw material' for adaptation.

Mathematical Patterns of Selection

Trait frequency changes follow predictable mathematical patterns. Under strong selection pressure, favorable trait frequency increases rapidly in early generations, then slows as it approaches fixation (100%). The Hardy-Weinberg principle describes the mathematical conditions under which trait frequencies remain stable — no selection, no mutation, no migration, large population, random mating.

Real-World Examples

The peppered moth in industrial England evolved from mostly light-colored to mostly dark-colored in just decades as pollution darkened tree bark. Darwin's finches on the Galapagos Islands show beak size shifts in response to drought years. Antibiotic-resistant bacteria demonstrate natural selection happening in days rather than years.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify environmental pressure, population diversity, favorable trait frequency, survival rate, and reproduction rate as system components.

E **Establish Relationships**
Students determine that environmental pressure negatively affects survival rate, population diversity positively affects favorable trait frequency, and favorable traits positively affect survival, which drives reproduction.

V **Visualize & Model**
Students build the natural selection model showing how environmental pressure and genetic diversity interact to drive population change.

E **Evaluate Outcomes**
Students compare low-diversity vs. high-diversity populations under identical environmental stress and analyze the mathematical difference in outcomes.

R **Revise & Extend**
Students add migration rate or mutation rate to explore how new genetic variation enters a population.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"If a deadly drought hit tomorrow, would every animal in the herd survive? Why not?"*

Watch For

Show images of diverse animal populations. Build suspense about who survives and why.

Slide 2: Learning Objectives

3 min

Say: *"Today we model one of biology's most powerful ideas — natural selection in action."*

Watch For

Have students read objectives. Pre-teach 'trait frequency' vs. 'genetic diversity.'

Slide 3: Big Question

3 min

Say: *"Can a species survive when the environment changes faster than it can adapt?"*

Watch For

Think-pair-share: What traits might help an animal survive a drought?

Slide 4: LEVER Framework

2 min

Say: *"We'll use the LEVER framework to build and test a model of natural selection."*

Watch For

Briefly preview each LEVER step.

Slide 5: Activity 1: Components

8 min

Say: *"What are the key parts of this system? Which ones can we control vs. which respond?"*

Watch For

Guide sorting of external vs. internal components. Emphasize that diversity is the raw material.

Slide 6: Activity 2: Connections

8 min

Say: *"Here's the key: pressure doesn't CREATE favorable traits — it REVEALS which ones matter."*

Watch For

Map positive and negative relationships. Focus on the feedback loop from reproduction back to trait frequency.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's test: will a diverse population survive what wipes out a uniform one?"*

Watch For

Students predict before running. Compare low vs. high diversity under identical pressure.

Slide 8: Discoveries

5 min

Say: *"What did the data reveal about diversity and survival? Were you surprised?"*

Watch For

Whole-class discussion. Connect to real examples like cheetah genetic bottleneck.

Slide 9: STEM Challenge

12 min

Say: *"You're conservation biologists. A species is declining — what's your evidence-based rescue plan?"*

Watch For

Distribute materials. Circulate and probe for model evidence in proposals.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Environmental Pressure, Population Diversity
- **INTERNAL:** Favorable Trait Frequency, Survival Rate, Reproduction Rate

Relationships

- **Environmental Pressure to Survival Rate:** NEGATIVE (-) — Harsher environmental conditions reduce the overall percentage of individuals that survive each generation.
- **Population Diversity to Favorable Trait Frequency:** POSITIVE (+) — Greater genetic diversity means a higher chance that some individuals carry traits suited to the current environment.
- **Favorable Trait Frequency to Survival Rate:** POSITIVE (+) — When more individuals carry advantageous traits, a greater percentage survives environmental challenges.
- **Survival Rate to Reproduction Rate:** POSITIVE (+) — More survivors means more individuals reproduce and pass on their traits to the next generation.
- **Reproduction Rate to Favorable Trait Frequency:** POSITIVE (+) — Surviving individuals with favorable traits reproduce, increasing the frequency of those traits in the next generation — this is natural selection in action.

Key Discoveries

- Populations with greater genetic diversity have more 'options' for natural selection to work with
- Environmental pressure doesn't create new traits — it selects for traits that already exist
- Low-diversity populations can crash rapidly under strong environmental pressure
- Natural selection is NOT random — it predictably favors traits that match the environment
- Reproduction rate amplifies the effect of selection by spreading favorable traits faster

Multiple Choice Post-Assessment Answer Key

1. In your model, a population with 80% genetic diversity survived a severe drought, while a population with 20% diversity collapsed. Which explanation is best supported by the model evidence?

Correct: B) The diverse population already contained individuals with drought-resistant traits that natural selection could favor

Correct! Natural selection acts on traits that already exist. The diverse population had a wider range of pre-existing traits, making it more likely that some individuals already carried drought-resistant variations. Selection favored those individuals.

2. Your model shows that Favorable Trait Frequency increased from 15% to 80% over 10 generations under high Environmental Pressure. Which component relationship explains this shift?

Correct: C) Individuals with favorable traits survived and reproduced at higher rates, passing those traits to offspring

Correct! This is natural selection in action. Individuals with favorable traits had higher survival rates, reproduced more, and passed those traits to the next generation. Over multiple generations, the frequency of favorable traits increased predictably.

3. A student claims that natural selection is random because mutations are random. Using evidence from your model, which response best addresses this claim?

Correct: C) The student is incorrect because while mutations are random, natural selection is directional, consistently favoring traits that match the environment

Correct! Mutations are random, but natural selection is not. In the model, increasing Environmental Pressure consistently and predictably reduced Survival Rate for individuals without favorable traits. Selection always favors traits that match the current environment.

4. Two populations face the same environmental pressure of 90%. Population A has 80% diversity and stabilizes at 70% favorable trait frequency. Population B has 20% diversity and crashes toward extinction. What does this model evidence reveal about conservation?

Correct: B) Maintaining genetic diversity in a population is critical because it provides the variation natural selection needs to drive adaptation

Correct! The model shows that genetic diversity is the raw material for natural selection. Without sufficient diversity, a population may lack individuals with traits suited to new conditions, leaving natural selection nothing to work with.

5. In your model, Reproduction Rate feeds back into Favorable Trait Frequency, creating a cycle. What does this feedback relationship represent in biological terms?

Correct: B) Surviving individuals with favorable traits reproduce and increase the proportion of favorable traits in the next generation

Correct! This feedback loop IS natural selection. Individuals with favorable traits survive, reproduce, and pass those traits to offspring. Each generation has a higher proportion of favorable traits, which increases survival rates further, creating a reinforcing cycle.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Survival of the fittest means the strongest survive"	Fitness in biology means reproductive success in a specific environment, not physical strength. A small, camouflaged lizard may be 'fitter' than a large, conspicuous one if predators are the main pressure.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Animals choose to adapt or evolve on purpose"	Individual organisms cannot choose to evolve. Natural selection acts on variation that already exists in the population. Giraffes didn't 'decide' to grow longer necks — individuals with slightly longer necks survived better and reproduced more.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Evolution always makes species better"	Natural selection optimizes for the CURRENT environment. If conditions change again, previously favorable traits may become harmful. Evolution doesn't have a direction or goal — it responds to present conditions only.